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Course: Blockchain

Research a blockchain identity project (uPort, Sovrin, Aadhaar)

Aadhaar Identity System Using Blockchain

The Aadhaar system is a national identity program in India that stores biometric and demographic data for residents. While Aadhaar has improved service delivery, its centralized database has raised concerns about privacy and security. To overcome these challenges, a blockchain-based Aadhaar Identity Management System (AIMS) has been proposed, integrating blockchain, smart contracts, and IPFS.

A major **strength** of blockchain-based Aadhaar is enhanced security through decentralization. Instead of storing all identity data in a single database, blockchain distributes records across multiple nodes, reducing the risk of large-scale data breaches. The system also introduces self-sovereign identity, allowing users to control when and how their Aadhaar data is shared. This increases user trust and transparency. In addition, immutability and cryptographic hashing ensure data integrity, making unauthorized changes easy to detect. Smart contracts further enable permission-based data sharing, ensuring that third parties access identity data only with user consent.

Despite these strengths, **several weaknesses** remain. Scalability is a major concern, as blockchain networks can process only a limited number of transactions per second, which may cause delays given Aadhaar's massive user base. The system is also technically complex, requiring knowledge of blockchain wallets and cryptographic mechanisms, which may limit adoption among non-technical users. Furthermore, blockchain immutability can amplify the impact of human errors, as incorrect data is difficult to correct once recorded. There is also a theoretical risk of 51% attacks, although such attacks are unlikely in large networks.

uPort Identity System

uPort is a decentralized identity platform built on the Ethereum blockchain, designed to support self-sovereign identity through mobile devices and smart contracts. Users create a digital identity linked to an Ethereum address and manage their credentials using a mobile application.

One of uPort's main **strengths** is strong user ownership and self-sovereignty. Users control their identity data through private keys and can decide what information to share and with whom. The system also benefits from decentralization, reducing dependence on centralized identity providers and lowering the risk of mass data breaches. Another advantage is selective disclosure, which allows users to reveal only the necessary information for verification, improving privacy. uPort is also flexible and interoperable, as it can integrate with decentralized applications on Ethereum.

However, uPort has notable **weaknesses**. Its heavy dependence on the Ethereum network means it inherits issues such as network congestion, high transaction fees, and slow

processing times. Key management is another challenge, as users are responsible for securing their private keys; losing them may result in loss of identity access. Additionally, usability and adoption remain concerns, since interacting with blockchain wallets and smart contracts can be difficult for users with limited digital literacy. While personal data is not stored on-chain, metadata visibility on Ethereum may still raise privacy concerns.

Sovrin Identity System

Sovrin is a blockchain-based identity network specifically designed for self-sovereign identity. Built on Hyperledger Indy, it uses a public-permissioned blockchain governed by trusted organizations known as stewards. Sovrin focuses on decentralized identifiers (DIDs) and verifiable credentials based on open standards.

A key **strength** of Sovrin is its strong privacy protection, supported by cryptographic techniques such as zero-knowledge proofs, which allow users to prove claims without revealing underlying data. Sovrin also emphasizes open standards and interoperability, enabling identity credentials to be used across different platforms and sectors. Another advantage is that sensitive personal data is stored off-chain, reducing exposure risks and supporting compliance with data protection regulations.

Despite these advantages, Sovrin also has limitations. Its permissioned governance model introduces some level of centralization, as only approved stewards can operate validator nodes. This may limit openness and raise concerns about power concentration. System complexity is another challenge, as managing decentralized identifiers and cryptographic credentials requires technical expertise. Scalability and global deployment also remain practical challenges, and like other self-sovereign systems, Sovrin places significant responsibility on users to manage and secure their identity credentials.

References:

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